

Machine Learning & Modern AI

Complete Step-by-Step Handwritten-Style Notes

ML → RAG → LLM → Embeddings → Vector DB → Fine-Tuning → MCP
→ Agentic AI

Study order: First understand ML foundations, then modern Generative AI concepts, then learn how RAG, MCP and Agentic AI work together.

1. Machine Learning (ML)

📌 Definition

Machine Learning is a branch of Artificial Intelligence in which a computer learns patterns from data and uses those patterns to make predictions or decisions, instead of being explicitly programmed for every case.

📌 Simple Example

If we give a model many examples of study hours and marks, the model can learn the relationship and predict marks for a new number of study hours.

📌 Key idea

Data → Learning/Training → Model → Prediction/Decision

2. Types of Machine Learning

📌 A. Supervised Learning

The training data contains both input features and a known target/output (labeled data).

📌 Common algorithms

Linear Regression, Logistic Regression, Decision Tree, Random Forest, SVM and KNN.

📌 B. Unsupervised Learning

The data has no target label. The algorithm discovers groups, patterns or structure by itself.

📌 Common algorithms

K-Means, DBSCAN and Hierarchical Clustering.

📌 C. Reinforcement Learning

An agent interacts with an environment, takes actions and learns from rewards or penalties.

📌 Examples

Chess AI, game-playing agents, robot navigation and self-driving systems.

3. Labeled Data, Classification and Regression

📌 Labeled data

Data where the expected output/target is already known. Example: Age, Experience → Salary.

📌 Classification

The target is a category/class. Examples: Spam/Not Spam, Pass/Fail, Disease/No Disease.

📌 Regression

The target is a continuous numeric value. Examples: salary, house price and temperature.

4. ML Lifecycle

📌 Step 1 — Problem Statement

Clearly define what must be predicted or decided, who needs the result and how success will be measured.

📌 Step 2 — Data Collection

Collect relevant data from CSV/Excel files, databases, APIs, sensors or other sources.

Step 3 — Data Cleaning

Handle missing values, duplicates, incorrect types, inconsistent values and obvious errors.

Step 4 — EDA

Exploratory Data Analysis helps understand distributions, relationships, outliers and correlations using statistics and visualizations.

Step 5 — Feature Engineering

Create useful input variables from raw data.

Step 6 — Feature Selection

Keep useful features and remove irrelevant or highly redundant features.

Step 7 — Train-Test Split

Separate data into training and testing sets. A common starting point is 80% train and 20% test.

Step 8 — Model Building

Choose an algorithm and train it using the training data.

Step 9 — Model Evaluation

Measure performance using suitable metrics such as Accuracy, Precision, Recall, F1, MSE, RMSE or R^2 .

Step 10 — Hyperparameter Tuning

Search for better hyperparameter values using techniques such as GridSearchCV or RandomizedSearchCV.

Step 11 — Deployment

Expose the trained model through an application/API or integrate it into a product.

Step 12 — Monitoring

Track model performance, data drift, failures, latency and business results after deployment.

5. Overfitting and Underfitting

Overfitting

The model learns training data too closely, including noise. Training performance is very high but performance on unseen data is poor.

How to reduce overfitting

Use more data, regularization, pruning, dropout where appropriate, cross-validation, simpler models or feature selection.

Underfitting

The model is too simple or insufficiently trained and fails to capture important patterns. Both training and test performance can be poor.

How to reduce underfitting

Use better features, a more suitable model, less restrictive regularization or better training.

6. MSE — Mean Squared Error

Definition

MSE is a regression loss/metric that calculates the average of the squared differences between actual and predicted values.

Interpretation

Lower MSE generally means predictions are closer to actual values. Because errors are squared, large errors receive more weight.

Example

Actual values = [10, 20], predictions = [12, 18]. Errors are [-2, 2], squared errors are [4, 4], so MSE = 4.

7. Encoding — Categorical Data to Numeric Data

Why encoding is needed

Most ML algorithms work with numerical representations, so categories often need to be converted into numbers.

Label Encoding

Maps categories to integers, for example Low=0, Medium=1, High=2. It is appropriate when the categories have a meaningful order or when the encoding is suitable for the chosen model.

One-Hot Encoding

Creates a separate binary column for each category. Example: Color=Red/Blue becomes Red and Blue columns.

Ordinal Encoding

Explicitly represents ordered categories, for example Low=1, Medium=2, High=3.

Important caution

Do not assign arbitrary numeric order to nominal categories when that would create a false sense of magnitude.

8. Feature Engineering

Definition

Feature engineering means transforming raw data into useful input features that help a model learn the target relationship.

Examples

DOB → Age; Date → Day/Month/Year; Price and Quantity → Total Amount; timestamp → hour/day/weekend.

Goal

Better features can improve model accuracy, generalization and interpretability.

9. ML Pipeline Workflow

Typical workflow

Raw Data → Cleaning → Encoding → Feature Engineering → Feature Selection → Scaling (if required) → Model Training → Evaluation → Prediction → Deployment.

Why use a pipeline

It keeps preprocessing and modeling together, makes training reproducible and helps avoid inconsistent preprocessing between training and prediction.

Data leakage warning

Fit transformations such as scaling/encoding on training data and apply the learned transformation to test/new data.

10. Pickle File and dump/load

📌 *Pickle*

Python's pickle module can serialize Python objects, including many trained ML models, into a file.

📌 *Pickle extension*

A common extension is .pkl, although the extension itself is not mandatory.

📌 *dump()*

Saves an object to a file: model/object → file.

📌 *load()*

Reads a saved object back: file → Python object.

📌 *Important security note*

Never unpickle untrusted files because pickle can execute arbitrary code during deserialization.

11. LLM — Large Language Model

📌 *Definition*

An LLM is a neural network trained on large amounts of text/data to learn language patterns and generate or transform text.

📌 *What it can do*

Answer questions, summarize, translate, extract information, write code and perform reasoning-like language tasks.

📌 *Basic flow*

Prompt → Tokenization → Model inference → Generated tokens → Text response.

📌 *Examples*

GPT-family models, Claude-family models, Gemini-family models and open-weight models such as Llama-family models.

12. Embedding

📌 *Definition*

An embedding is a numerical vector representation of text, an image, audio or another object that captures useful semantic information.

📌 *Simple idea*

Similar meanings tend to have vectors that are closer together in embedding space.

📌 *Example*

'car' and 'automobile' should generally have more similar embeddings than 'car' and 'banana'.

📌 *Uses*

Semantic search, RAG, recommendation systems, clustering, duplicate detection and retrieval.

13. Vector Database

Definition

A vector database stores embeddings and supports similarity search over those vectors.

Why it is needed

Normal keyword search may miss meaning. Vector similarity can retrieve information that is semantically related even when exact words differ.

Common concepts

Embedding, vector index, metadata, similarity metric, top-k retrieval and filtering.

Examples

Pinecone, Weaviate, Milvus, Qdrant and vector capabilities in several general-purpose databases.

14. RAG — Retrieval-Augmented Generation

Definition

RAG combines retrieval from an external knowledge source with an LLM's generation step.

Why RAG

The model can use current, private or domain-specific information without retraining the entire model every time the source documents change.

Step 1

Collect documents such as PDFs, web pages or internal files.

Step 2

Split documents into meaningful chunks.

Step 3

Create embeddings for the chunks.

Step 4

Store embeddings and metadata in a vector database.

Step 5

Convert the user's query into an embedding.

Step 6

Retrieve the most relevant chunks using similarity search.

Step 7

Put the retrieved context into the LLM prompt.

Step 8

LLM generates an answer grounded in the retrieved context.

Flow

Documents → Chunking → Embeddings → Vector DB → Query → Retrieval → Context + Prompt → LLM → Answer

15. Fine-Tuning

Definition

Fine-tuning means continuing the training of a pretrained model on a smaller, task/domain-specific dataset so its behavior becomes better suited to that task.

📌 *When useful*

Consistent style, classification, extraction, domain behavior, structured outputs or specialized task behavior.

📌 *Basic flow*

Pretrained model → Prepare high-quality examples → Fine-tune → Evaluate → Deploy.

📌 *RAG vs Fine-tuning*

RAG mainly gives the model external information at inference time. Fine-tuning changes model behavior/weights through additional training.

📌 *Practical rule*

If the main problem is changing or private knowledge, RAG is often a better first choice. If the problem is behavior or task style, fine-tuning may be more appropriate.

16. MCP Protocol

📌 *Definition*

MCP (Model Context Protocol) is an open protocol for connecting AI applications/models to external tools and context in a standardized way.

📌 *Problem it solves*

Without a common protocol, every AI application may need a custom integration for each tool or data source. MCP provides a standard interface.

📌 *Core idea*

AI application/host ↔ MCP client ↔ MCP server ↔ external tools/resources.

📌 *MCP Server*

An MCP server exposes capabilities such as tools, resources or prompts to an MCP client. The server may connect to files, databases, APIs, business systems or other services.

📌 *Example*

An AI assistant can use an MCP server that exposes a database-search tool. The assistant requests the tool, the server performs the operation and returns structured results.

📌 *Important distinction*

MCP is an integration protocol. It is not itself an LLM, RAG system or vector database.

17. Agentic AI

📌 *Definition*

Agentic AI refers to AI systems that can pursue a goal through multiple steps, choose actions, use tools, observe results and continue until the task is completed or stopped.

📌 *Traditional chatbot*

User asks → model answers.

📌 *Agentic system*

Goal → plan/reason → choose tool/action → execute → observe result → update state → next action → final result.

📌 *Key components*

LLM, tools, memory/state, planning or workflow logic, observation/feedback and safety/permission controls.

Example

A travel agent can understand a trip goal, search available options, compare them, ask for missing information and prepare a final itinerary.

MCP connection

MCP can provide a standardized way for an agentic AI application to discover and use external tools or resources.

18. How These Concepts Connect

Big picture

ML is the broader field of learning from data. Deep Learning uses neural networks. LLMs are large neural models specialized in language and multimodal tasks.

Modern AI application stack

User → Agent/LLM → Tools or MCP → RAG Retriever → Vector DB/Knowledge Sources → External APIs/Systems → Response.

Important distinction

Embeddings represent information numerically; vector databases store/search those representations; RAG retrieves useful context; LLM generates language; fine-tuning changes model behavior; MCP standardizes connections to external capabilities; agentic AI orchestrates multi-step actions.

19. Interview Quick Revision

ML

Learning patterns from data for prediction/decision making.

Supervised

Uses labeled data.

Unsupervised

Finds patterns without target labels.

Classification

Predicts a class/category.

Regression

Predicts a continuous number.

Overfitting

High training performance, poor generalization.

Underfitting

Model is too simple and learns insufficiently.

MSE

Average squared prediction error.

Encoding

Converts categorical values into model-usable numeric representation.

📌 *Feature Engineering*

Creates useful features from raw data.

📌 *Pickle*

Serializes Python objects; common for saving ML models.

📌 *LLM*

Large neural language model trained on large datasets.

📌 *Embedding*

Numerical semantic representation.

📌 *Vector DB*

Stores/searches vectors using similarity.

📌 *RAG*

Retrieves relevant external context before generation.

📌 *Fine-tuning*

Additional training to adapt model behavior/task performance.

📌 *MCP*

Standard protocol for connecting AI applications to external tools/context.

📌 *MCP Server*

Exposes tools/resources/prompts through MCP.

📌 *Agentic AI*

Goal-driven AI that can plan/use tools/observe and take multiple steps.